Why Do Restaurants Centralize Food Production? Identifying Hidden Production Practices Through Cross-Market Signals

Eric Wu[1, †], Vincent Lau[2, †], Kane Wu[3], Colin Sheng[4] *and* Uchenna Cyril Eze[5, *]

[1] Faculty of Business and Management, Beijing Normal–Hong Kong Baptist University.
[2] School of Data Science, The Chinese University of Hong Kong, Shenzhen.
[3] Faculty of Business and Management, Beijing Normal–Hong Kong Baptist University.
[4] Faculty of Computer and Mathematical Sciences, The Hong Kong Polytechnic University.
[5] Faculty of Business and Management, Beijing Normal–Hong Kong Baptist University.

[†] These authors contributed to this work equally.
* Correspondence: Uchenna Cyril Eze, uchennaeze@bnbu.edu.cn

Working Paper
October, 2026

Abstract

The growing centralization of food production has transformed restaurant operations, but its adoption is difficult to investigate because production arrangements are often unobservable to outsiders. This study examines the economic and institutional factors associated with restaurants' reliance on centralized production using a latent-variable approach. Building on the task-based framework, we propose that transferring culinary tasks from individual outlets to central kitchens or upstream suppliers reduces the relative demand for on-site culinary skills, making the chef-to-server wage ratio a potential indicator of hidden production arrangements. Using data from 311 restaurants across five Chinese cities, we estimate a Multiple Indicators Multiple Causes (MIMIC) model with the wage ratio and the proportion of negative reviews mentioning prepared meals as reflective indicators. Results show that centralized production reliance is negatively associated with the wage ratio and positively associated with commercial rent, chain scale, and restrictions on open-flame cooking. Robustness tests support these patterns. External validation further shows that restaurants with publicly documented centralized production have a significantly lower mean wage ratio than restaurants without such disclosures. These findings suggest that centralization reflects not only efficiency considerations but also local operating costs, institutional constraints, and the standardization requirements of chain expansion. The study also points out that opaque production practices in one market may be inferred from observable traces in an adjacent market.

Keywords: Centralize Food Production; Labor-market Signals; Restaurant; MIMIC Model; China; Task-based Framework.

JEL Classification: D82; J31; L83.

---

1 Introduction

The centralization of food production has become the most structural transformation in the restaurant industry over the past few decades. Under the centralized production mode, major production activities – ingredient procurement, processing, cooking and packaging are relocated to a centralized production facility then the semi-finished or finished food are shipped to branches for final preparation and service (Zhang, 2025). Evidence in the United State illustrated a 2.2% compound annual growth rate (CAGR) of

chain restaurant from 2021 to 2024 (IBISWorld, 2024). As restaurant chains expand, the need to control operating costs and maintain consistent food quality becomes increasingly important, making centralized food production an attractive operational strategy (Zhang, 2025).

In the context of the trend of centralized production, the catering industry in the Chinese mainland is a typical research object. China has witnessed explosive growth of the centralized food production industry: the market size of centralized food production have exceeded 500 billion RMB by 2023 and industry estimates suggest that prepared meals[1] had already penetrated approximately 12% of restaurant food procurement in China by 2021 (Tungee Research Institute, 2022). Projections suggest continued double-digit annual growth rates driven by government policy support, rising urbanization, increasing labor costs, and the expansion of chain-operated restaurant brands (Liu et al., 2022).

However, empirical investigations into the adoption and diffusion of centralized food production remain constrained by a fundamental measurement challenge: whether a given restaurant has adopted this production mode is extremely difficult to observe (Xue et al., 2025). The observability issue is divided into two interrelated factors. First, restaurant kitchens have traditionally been organized as back-of-house spaces, physically separated from dining areas (Byun and Jang, 2018). Consequently, the whole process is finished under the structure of inner production and is invisible to external observers. Second, Shan et al. (2026) states that the probability for restaurant to disclose their production mode depends on traceability costs or penalties. In the Chinese context, consumers have a very strong aversion to dishes not freshly prepared (Xue et al., 2025). Thus, Chinese consumers react negatively to foods prepared in a central setting (see Section 2.2). This attitudinal backdrop leads to a weak disclosure intention of restaurants that adopt centralized production mode.

Existing research has provided important insights into the economic rationale and operational implications of centralized food production. For instance, Nettles et al. (1997) finds that projected labor costs and return on investment are important considerations in selecting centralized food production systems in institutional food service operations. Rodgers (2005) highlights the importance of operational efficiency, production capacity, and cost considerations in food production system selection. More recently, Zhang (2025) develops an analytical model showing that investment in centralized production facilities can improve food quality consistency. However, systematic empirical evidence on the factors associated with restaurants' adoption of centralized production remains limited. In particular, little is known about how restaurant-level economic and institutional conditions shape the degree of production centralization.

To address this gap, we have to solve the observation problem of restaurants' production arrangements. When a restaurant centralizes food production, it inevitably outsources some of the cooking tasks to upstream factories or central kitchens (Deng, 2023). We applied Acemoglu and Autor's (2011) tasks based model to illustrate that such outsourcing can reduce the chef-to-server wage ratio (see Section 3.4). Additionally, restaurants' centralized food production can also be inferred from consumers' feedback, since a number of Chinese consumers are sensitive to the food produced by central kitchen while leaving negative reviews (see Section 2.2). These indicators, derived from underlying economic and social mechanisms, provide indirect measures of restaurants' unobservable reliance on centralized food production.

To investigate the factors associated with restaurants' adoption of centralized food production while validating the proposed indicators, Multiple Indicators Multiple Causes (MIMIC) model is applied to this study. The MIMIC model is a structural-equation framework that links observable factors that influence an unobserved latent variable to multiple observable indicators of that latent variable (Jöreskog and Goldberger, 1975; Chen and Jiang, 2019). This approach has been applied in business and hospitality management research to estimate unobserved managerial characteristics using operational data (Tsionas et al., 2022) and to examine how observable variables are associated with latent constructs of hotel guests' behavioral responses (Chen and Jiang, 2019). In this study, the restaurants' reliance on centralized food production is the unobserved latent variable. Using data from 311 restaurants across five major Chinese

[1]The relationship between prepared meals and centralized food production will be discussed in Section 2.2

cities, we find that higher commercial rent, larger chain scale and open fire regulations are positively associated with restaurants' food production centralization. The external validation using restaurants with publicly disclosed central kitchen further supports the validity of the measure.

This study makes three major contributions. First, to the best of our knowledge, this study is the first to empirically examine the operational factors associated with restaurant production centralization. Second, it contributes to research on introducing a scalable approach to measuring hidden operational configurations, providing valid indicators for future researchers in studying restaurants' centralized food production. Third, it shifts the perspective of task-based research. Whereas existing studies address how observed offshoring affects relative wages, this study address what observed wage structures can reveal about otherwise unobservable task reallocation, hereby extending the framework to domestic outsourcing and centralized production.

2 Literature Review and Theoretical Background

2.1 Restaurant Production Organization and Back-of-House Operations

Different from other retailing industry, restaurant operations are commonly divided into front-of-house (FOH) and back-of-house (BOH) areas (O'Neill et al., 2024). While FOH primarily handles customer-facing services, BOH is responsible for food production, including storage, preparation, and cooking (Lee et al., 2024). Traditionally, these two functions operate within the same restaurant, requiring coordination between service delivery and food production. Centralized food production extends this organizational separation by relocating part of the BOH activities to centralized facilities that supply multiple restaurant outlets (Zhang, 2025). Under this arrangement, FOH services and certain BOH tasks remain at individual outlets, while major food-production activities are performed centrally. This restructuring enables economies of scale and production standardization but introduces additional coordination requirements, including logistics and inventory management (Rodriguez et al., 2025). Thus, centralized food production can be understood as an extension of the traditional FOH–BOH division, in which food production and customer-facing services become increasingly separated across locations.

2.2 Consumer Controversies and Perceptions of Centralized Food Production

According to Xue et al. (2025), prepared meals are defined as pre-packaged finished or semi-finished meals made from agricultural, livestock, poultry, and aquatic products through pre-processing or pre-cooking in the central kitchen. In restaurant operations, prepared meals and centralized food production represent two closely related aspects of the same production system. While the former refers to food products prepared in advance, the latter describes the organizational mode in which food preparation is shifted from individual restaurant outlets to centralized processing facilities (Jia et al., 2024; Zhang, 2025). Accordingly, both the use of externally sourced prepared meals and the operation of in-house central kitchens can be understood as forms of centralized food production. In this study, prepared meals are conceptualized as products of centralized food production, regardless of whether they are produced by internal supplier or sourced from external suppliers[2].

The application of centralized food production in China's restaurant industry has generated substantial public controversy, sentiment analysis of 200,993 social media comments related to prepared meals disclosed that approximately 64% exhibited negative emotion, reflecting public opposition to prepared dishes (Shu et al., 2024). The interview conducted by Xue et al. (2025) also shows that a number of

[2]Since 2024, Chinese regulatory definitions have distinguished central kitchens from other prepared-food producers by defining central kitchens as centralized food-processing facilities established by catering service providers to process and distribute food to their own outlets (State Administration for Market Regulation, 2024). Despite this institutional distinction, both central-kitchen production and externally sourced prepared meals involve the transfer of food preparation from individual restaurant kitchens to centralized production facilities and thus belong to the broader mode of centralized food production.

Chinese consumers regard prepared meals as "inferior" to freshly cooked meals, and should charge a lower price, but not pretend to be freshly cooked and charge a relatively high price.

Concerns about prepared meals originate from various aspects. The first concern is related to price. Using prepared meals is an effective way to reduce cost (Zhou, 2021). However, the lack of prior information may cause consumers to pay prices for prepared meals comparable to those charged for freshly cooked meals. Secondly, China has a history of food safety issues, especially concerning food products manufactured through industrialized production processes: Melamine-Tainted Milk Scandal in 2008 [3], Gutter Oil Scandal from 2011 to 2012 [4], Preserved Vegetable and Pork Prepared-Dish Scandal in 2024 [5], and Lead-Contaminated Food Incident in 2025 [6]. These events often involved improper actions by the supervisory authorities, which have led to a lack of social trust. Though prepared meals are not necessarily unhealthy compared with fresh meals, Chinese consumers often associate prepared meals with past food safety scandals (Xue et al., 2025). Thirdly, the taste of prepared meals is also important. The Chinese term *Wok Hei* (the breathe of the *wok*) [7], is an indispensable aspect of judging the taste of a Chinese dish. Prepared meals are sometimes considered "lacking of *Wok Hei*" [8]. Chinese consumers regard *Wok Hei* as an important component of the dining-out experience. When restaurants serve prepared meals lacking *Wok Hei*, consumers may feel frustrated (Xue et al., 2025).

2.3 The Impact of Production Mode Change on Labor Structure

This study draws on the task-based view of production model change to explain why labor-market information may reveal hidden production practices in restaurants.

Acemoglu and Autor (2011) define task as a unit of work activity that produces output, and the output including both service and goods. Different from the canonical skill-based model, treating a job as a fixed bundle of skills, which is insufficiently nuanced to account for the rich relationships among skills, tasks and technologies, the task-based framework distinguishes between workers' skills and the tasks they perform (Autor et al., 2003). Under task-based framework, workers apply their skill endowments to tasks in exchange for wages, and skills applied in tasks produce output (Acemoglu and Autor, 2011). This distinction is important because technological or organizational change may not directly alter workers' skills, but it can change which tasks remain inside the firm, which tasks are simplified, and which tasks are transferred to machines, suppliers, or external production sites. Sampson (2021) based on this task-based framework develops a professional task-automation framework that shows how individual tasks within a given job can be enhanced or disrupted by automation in very different ways.

Acemoglu and Autor's task-based framework is also especially useful for this study because it explains how production can be reorganized without being immediately visible. In task-based framework, the labor market are not only simply changed by the variation of productivity, but also changed by the varying allocation of tasks across different skill workers (Acemoglu and Autor, 2011). Acemoglu and Autor (2011) assume that there are three types of skills – low, medium and high – and each worker is endowed with one of these types of skills. In this setting, intermediate tasks are often performed by medium-skilled

[3] Melamine was illegally added to diluted milk to falsify protein-test results. See Reuters, "Factbox: What Is Melamine, and Why Add It to Milk?," January 22, 2009, Reuters.

[4] Waste oil collected from restaurant grease traps, discarded animal products, and other unsanitary sources was illegally processed and resold as edible cooking oil through underground supply chains. See Reuters, "China Arrests 52 over Cooking Oil Scandal," December 20, 2011, Reuters.

[5] Scandal directly related to prepared meals. See CCTV Finance, "Lymph Nodes and Glandular Tissue Clearly Visible: Disturbing Production Conditions behind Packaged *Meicai Kourou*," March 15, 2024, CCTV News (in Chinese; title translated by the author).

[6] A kindergarten allegedly added non-food-grade coloring pigments containing high levels of lead to cakes and other foods. See Xinhua, "Kindergarten Staff Arrested after 247 Children Poisoned with Lead in NW China," July 20, 2025, Xinhua.

[7] The term *wok* refers to a round-bottomed Chinese cooking pan traditionally used for stir-frying, deep-frying, steaming, braising, and other high-heat cooking techniques. *Wok Hei* refers to "the energy a wok bestows upon a stir-fry, giving foods a unique smoky flavor and aroma" (López-Alt, 2012)

[8] Further disscusion on why prepared meals sometimes lack *Wok Hei* can be found in Section 3.3.

workers and are more susceptible to automation or relocation than many low-end in-person service tasks and high-end managerial tasks (Acemoglu and Autor, 2011).

This logic can be applied to restaurant production. A restaurant does not produce meals through a single homogeneous task, but through a series of differentiated tasks, including food preparation, cooking, plating, serving, cleaning (Göde and Ekergil, 2020).

According to Blinder (2007) and Blinder and Krueger (2008), any job that does not need to be done in person (i.e., face-to-face) can ultimately be outsourced, regardless of whether its primary tasks are abstract, routine, or manual. For customer-service-related tasks, although they do not require highly specialized skills, they do require situational adaptability, visual and linguistic recognition, and face-to-face interaction during execution, which makes them non-transferable outside the restaurant setting (Acemoglu and Autor, 2011). For tasks such as cooking or preparation procedures, can be centralized in central kitchen. Once semi-finished or finished food products are produced, they are distributed to restaurants, where the remaining on-site culinary tasks may be reduced to reheating, plating, or final-stage cooking (Deng, 2023).

Therefore, the application of centralized food production can be interpreted as a form of task reallocation. In a traditional freshly cooked model, a substantial share of intermediate tasks is performed in the restaurant kitchen. Chefs are required to perform cooking-related tasks on site, including preparation, timing, heat control, flavor adjustment, and quality stabilization(China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute, 2023). These tasks generate demand for culinary labor, especially for chefs with medium skill level who are directly involved in dish production.

However, as discussed by Robinson and Barron (2007), technological applications and outsourced food production can contribute to the deskilling and standardization of culinary work. When a restaurant relies more heavily on centralized production, some culinary tasks are transferred away from the on-site kitchen, reducing its reliance on skilled culinary labor. The restaurant may still provide similar dishes to consumers, but the allocation of production tasks behind those dishes has changed. A chef may retain the same culinary ability, but if fewer skill-intensive cooking tasks remain within the restaurant, the restaurant has less incentive to pay a high wage premium for that ability. In contrary, the tasks for server will probably remain same after the adoption of prepared meals or central kitchen production mode, so their will likely remain stable, leading to a smaller wage difference between server and chef.

3 Hypotheses Development

It is difficult for researchers to directly observe restaurants' reliance on centralized food production. However, theoretical discussion on factors that influence the reliance on centralized food production and its indicators is possible. The following analysis establishes three factors associated with centralized food production and develops two indicators to measure this hidden production practice, laying a solid foundation for the subsequent empirical analysis.

3.1 Commercial Rent and Production Centralization

As we have discussed in Section 2.1, restaurant operations are commonly divided into front-of-house (FOH) and back-of-house (BOH) areas. Higher commercial rent increases the opportunity cost of allocating floor space to back-of-house operations. Additional front-of-house space can often be monetized relatively directly by accommodating more tables and customers, whereas back-of-house space contributes to revenue indirectly by supporting food production. This does not imply that kitchen space does not create value. Rather, compared with dining space, kitchen space cannot usually expand customer capacity simply by accommodating additional seats. This spatial trade-off is particularly important in the restaurant industry because customer demand is often concentrated within relatively short meal periods (Wang et al., 2018). During peak dining hours, seating capacity and operational throughput therefore become

especially valuable.

Traditional on-site cooking process requires restaurants to maintain sufficient and functionally differentiated back-of-house space for storing ingredients, handling unprocessed raw materials, preparing and cooking food, separating finished products from production waste, and preventing interference or contamination across different production processes (Flessas et al., 2015). By contrast, restaurants that adopts centralized production can relocate some of these activities to upstream production facilities. Consequently, restaurants with centralized food production can serve the same quantity or even more orders with a smaller kitchen, enjoying lower rental costs and increasing revenue (Deng, 2023). Industry evidence illustrates the potential magnitude of this cost-saving mechanism. According to Zhou (2021), the rental cost of a typical Chinese dish allocated to each order decreased by approximately 40% when on-site cooking was replaced with prepared meals (see Table 1). Therefore, restaurants facing higher rental cost may have a higher incentive to cut the cost by adopting centralized production.

TABLE 1. Cost & Profit Comparison of a Typical Chinese Dish

Category	Freshly Cooked	Prepared Meals	Percentage Change
Packaging cost	¥ 2.00	¥ 2.00	
Platform promotional fees	2.40	2.40	
Platform commission	5.20	5.20	
Ingredient cost	6.50	7.50	+ 15.38%
Labor cost	1.60	0.80	– 50.00%
Rental cost	1.00	0.60	– 40.00%
Utilities and miscellaneous expenses	0.40	0.20	– 50.00%
Selling price per serving	20.00	20.00	
Profit per serving	0.90	1.30	+ 44.44%

Note: Percentage change is calculated as (Prepared Meals – Freshly Cooked)/Freshly Cooked.
Source: Authors' compilation based on Zhou (2021).

Accordingly, commercial rent is incorporated into the MIMIC model as a theoretically motivated cause of latent centralized food production reliance.

Hypothesis 1. *A restaurant's latent reliance on production centralization is positively affected by higher local commercial rent*

3.2 Chain Scale and Production Centralization

For large chain restaurants, the core challenge is maintaining consistency while expanding geographically (Bradach, 1998). As the result, compared to individual restaurants, large chain restaurants has greater pressure to ensure their consistency and replicability for further expansion. Zhang (2025) states that after introducing central kitchen production mode, firms can improve quality consistency across outlets. According to China Chain Store & Franchise Association and China Renaissance (2022), the trend toward restaurant chain expansion further promotes the application of centralized food production.

Consequently, the central kitchen model meets restaurant chains' cost reduction and efficiency improvement needs (Deng, 2023). Compared to restaurants with fewer stores or independent restaurants, large restaurant chains appears to have a stronger incentive to adopt centralized food production.

Accordingly, number of chain stores is incorporated into the MIMIC model as a theoretically motivated cause of latent prepared-meal reliance.

Hypothesis 2. *A restaurant's latent reliance on production centralization is positively affected by the number of chain stores.*

3.3 Regulatory Constraints and Production Centralization

According to the regulations for catering areas in large commercial complexes (Ministry of Emergency Management of the People's Republic of China, 2023, Section 9.2b), restaurants located underground or inside large commercial complexes face stricter fire-safety and facility-layout constraint. Catering establishments shall not use liquefied petroleum gas (LPG) or Class A or Class B liquid fuels. Additionally, kitchen areas using open fire should be arranged against the exterior wall (Ministry of Emergency Management of the People's Republic of China, 2023, Section 9.2e). Facing strict fire compliance pressure, restaurants located in large commercial complexes are more likely to give up open flame cooking.

As mentioned before, Chinese dishes emphasize *Wok Hei*, which to some extent relies on open-fire cooking. Relevant research suggest that the formation of *Wok Hei* is closely associate with the browning enabled by the Maillard reaction (MR) (Ko and Hu, 2020). Traditional high-temperature dry-heat cooking methods, such as open-flame cooking, may be more conducive to surface dehydration, temperature increase, and browning, thereby contributing to flavor development (Purlis, 2010). By contrast, studies on microwave suggest that insufficient browning and flavor development are typical limitations of microwave heating (Sumnu, 2001; Dong et al., 2022). This may be the reason why some Chinese consumers claim that prepared meals that are simply reheated by microwave taste inferior to freshly cooked meals.

Additionally, the cold-chain transportation of centralized food production may further accelerate the loss of desirable volatile compounds and the production of off-flavor compounds (Yu et al., 2020). Complaints on "gas stove ban" by Chinese chefs occur frequently because they think techniques like "stir-frying" need open flame for rapid and high-intensity heat [9].

Therefore, it is reasonable to make such inference that if gas is banned in large commercial complexes, and if induction cooking cannot provide satisfactory flavor, restaurants may rather apply centralized food production to cut the expense on skilled chefs since their skills no longer generate much value for consumer. Incidents in which restaurants adopted centralized production and dismissed chefs due to fire-safety regulations have indeed been reported by the Chinese media[10].

Accordingly, whether there is open fire regulation is incorporated into the MIMIC model as a theoretically motivated cause of latent centralized food production reliance.

Hypothesis 3. *A restaurant's latent reliance on production centralization is positively affected by open fire ban for restaurants located in large commercial complexes in the interior.*

3.4 Production Centralization and Labor-Market Signal

To establish the theoretical basis for the proposed wage-ratio indicator, we adapt the task-based offshoring framework of Acemoglu and Autor (2011) to the restaurant context. The detailed theoretical derivation of this mechanism, including the task allocation structure and equilibrium wage implications, is provided in **Supplementary Material S1**. According to the result of theoretical derivation, the relative wage structure is affected as follows:

$$WageRatio = \frac{w_M}{w_L} \downarrow \tag{1}$$

In Equation (1), w_M represents the wage of middle-level chef and w_L represents the wage of server. This mechanism is consistent with Acemoglu and Autor (2011) implications of offshoring on the structure of wages, providing a task-based theoretical rationale for the predicted direction of why the chef-to-server

[9]For media reports, see Jenn Harris, "The End of Korean BBQ in L.A.? What the Gas Stove Ban Means for Your Fave Restaurants," *Los Angeles Times*, June 2, 2022, https://www.latimes.com/food/story/2022-06-02/gas-stove-ban-chinese-korean-bbq-electric-new-buildings-restaurants-future; Naoki Nitta, "Would Giving Up Gas Stoves Threaten the Art of Chinese Cooking?," *San Francisco Chronicle*, January 30, 2023, https://www.sfchronicle.com/food/article/gas-stove-wok-chinese-17742332.php; These reports quote Chinese or Asian-restaurant chefs and owners who argue that gas and open flame is important for wok cooking, high-heat stir-frying, and the production of wok hei or smoky flavor.

[10]See Hu Kefei, "The Chef Disappears from the Back Kitchen," *China Newsweek*, January 10, 2022, https://www.inewsweek.cn/viewpoint/2022-01-10/14884.shtml.

wage ratio can decline even in the absence of changes in individual skill distributions, as it reflects a contraction in the set of tasks performed by chefs rather than a change in their underlying ability. However, this economic derivation just provides a theoretical prediction, the further validation of this indicator will be conducted in Section 5.2.4.

This implication also explains why this study uses the chef-to-server wage ratio rather than the absolute chef wage. Since the adoption of centralized food production externalizes part of the in-house culinary task interval previously performed by middle-skill chefs while operating as a productivity-enhancing technology, as standardized upstream production can reduce unit costs, increase operational efficiency, and raise the effective productivity of restaurant production (China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute, 2023). Therefore, the direction of the absolute chef wage is theoretically ambiguous.

By contrast, the chef-to-server wage ratio is more directly linked to the relative task structure inside the restaurant. When centralized food production relocates a substantial part of cooking, seasoning, heat control, and food-preparation tasks to central kitchens or upstream suppliers, the remaining in-store culinary work becomes closer to simple finishing, reheating, and plating. In the limiting case, the task requirements of in-store kitchen workers become increasingly similar to those of low-skill restaurant workers. According to the law of one price for skills, workers performing tasks with similar skill requirements should receive similar wages in equilibrium (Acemoglu and Autor, 2011). Hence, as the in-house chef task set contracts and becomes less distinct from low-skill restaurant tasks, the chef-to-server wage ratio should move closer to one.

The wage ratio also reduces the influence of local labor-market and demand-side conditions. Absolute chef wages may vary substantially across cities, neighborhoods, and restaurant segments because of differences in local income levels, consumption capacity, rent, and general wage levels (Combes et al., 2008). A given chef wage level therefore cannot by itself identify whether a restaurant relies heavily on prepared meals. The chef-to-server wage ratio normalizes chef wages by the wage of another occupation within the same local restaurant labor market, making it a more appropriate proxy for changes in the relative demand for in-house culinary skills.

Hence, the relationship of the reliance on centralized food production and the chef-to-server wage ratio is stated as follows:

Hypothesis 4. *The chef-to-server wage ratio of a restaurants is negatively associated with its reliance on production centralization.*

3.5 Production Centralization and Consumer Signals

As we have discussed in Section 2.2, Chinese consumers are highly sensitive to restaurants' usage of prepare meals, which provides us a signal from consumers' reaction to identify restaurants' reliance on centralized production. Although no scientific experiments can prove that consumers can accurately determine the difference between prepared meals and freshly cooked meals, some studies have shown that consumers who have already dined in the restaurant before can infer whether the restaurant uses prepared meals secretly by using the degree of processing (Hässig et al., 2023), sensory perception of food (Bai et al., 2019) or unusual serving speed (Xue et al., 2025) as signals.

However, simply focusing on the total number of reviews mentioning prepared meals is not methodologically appropriate. Restaurants with a large number of reviews may naturally receive a greater number of reviews mentioning prepared meals. However, this does not necessarily indicate that they rely more on centralized production than restaurants that receive fewer such reviews. Therefore, the absolute number of reviews mentioning prepared meals alone may not accurately reflect restaurants' reliance on centralized food production.

Therefore, we define the $CommentRatio$ for restaurant as

$$CommentRatio = \frac{N^{PM}}{N} \tag{2}$$

In Equation (2), N^{PM} denotes the number of negative reviews that explicitly mention prepared meals or related production practices (Central Kitchen), and N denotes the total number of negative reviews received by the restaurant. A higher Comment Ratio is consequently expected to reflect greater reliance on centralized food production and should load positively on the latent centralized-production-reliance construct.

This construction measures the proportion of consumer dissatisfaction that is specifically attributed to prepared-meal production. It has three advantages over a raw review count. First, it normalizes for differences in restaurant popularity and total review exposure. Second, it distinguishes prepared-meal-related dissatisfaction from general dissatisfaction caused by service, price, sanitation, waiting time, or the dining environment. Third, because the numerator is a subset of the negative-review denominator, both are exposed to more comparable review-generation, fraud-detection, complaint, and moderation processes. The ratio may therefore partially absorb restaurant-level differences in the visibility and retention of negative reviews. Moreover, as noted by Jöreskog and Goldberger (1975), in MIMIC model, indicators are conditionally independent given the latent variable. Since $WageRatio$ and $CommentRatio$ arise from substantively distinct information-generating processes: the former reflects labor-market outcomes, whereas the latter reflects consumers' dining experiences. Their joint use therefore provides two independent manifestations through which the proposed latent construct can be evaluated, rather than relying on multiple measures generated by the same underlying information source.

Nevertheless, the indicator is not a direct verification of production practices. Consumer misclassifications and fraudulent reviews may still generate measurement error. We therefore use the $CommentRatio$ only as an auxiliary reflective indicator rather than as a stand-alone classification criterion. In the robustness analysis, we additionally impose minimum-review thresholds to reduce instability arising from small denominators and change $CommentRatio$ into other form to show that its validity does not rely on its specific formulation.

Hypothesis 5. *The prepared-meal-related negative reviews are positively associated with its reliance on production centralization.*

4 Research Method

This study adopts a theory-driven latent-variable approach to investigate restaurants' unobservable reliance on centralized food production. Deriving from Acemoglu and Autor (2011) task-based framework, we suppose that $WageRatio$ can be a reasonable indicator to estimate restaurant's reliance on centralized food production, since the reallocation of task caused by centralized production can lead to a change in wage structure. Additionally, because of Chinese customers' sensitiveness to prepared food, we proposed that $CommentRatio$ can also be a indicator to reflect restaurant's extent of centralized production (see section 3).

Then, a Multiple Indicators Multiple Causes (MIMIC) model is employed to examine whether the observed relationships are consistent with the theoretically specified common-factor structure. The MIMIC model is for inferring latent variable through the systematic relationships among observable variables (Jöreskog and Goldberger, 1975). The model jointly estimates a set of structural and measurement equations in which the latent variable is linked to both its observed factors and its observable indicators (Frey and Weck-Hanneman, 1984). In our setting, the latent construct is a restaurant's degree of reliance on centralized food production. It is modeled as being influenced by a set of theoretically motivated factors and reflected in observable indicators.

The MIMIC model consists of two sets of equation:

$$\mathbf{y} = \lambda\eta + \varepsilon \tag{3}$$

$$\eta = \gamma'\mathbf{x} + \zeta \tag{4}$$

where Equation (3) is the measurement model for the latent variable η and Equation (4) is the structural equation equation for the latent variable η. In our study, $\mathbf{y}$ is a column vector of 2 indicators of the single latent variable, and $\mathbf{x}$ is a vector of 3 factors of η. In the Equation (4), the latent performance is assumed to be caused by the vector of explanatory variable $\mathbf{x}$. ε refers to a vector of zero mean (2×1) measurement error variables associated with the indicators, while ζ is a zero mean scalar structural error that captures unmodeled variables affecting η and the measurement error associated with it. It is assumed that ζ and all the elements of ε are mutually uncorrelated.

According to Jöreskog and Goldberger (1975), for a standard scalar MIMIC model, identification requires at least two reflective indicators with uncorrelated measurement errors and at least one observed causal variable. In our MIMIC model, we have 3 factors (observed causal variable) and 2 reflective indicators, which lives up to its identification requirements.

This study assume that a restaurant's reliance on centralized production is manifested through labor market signals and consumer-market signals. We use $WageRatio$ in Equation (1) and $CommentRatio$ in Equation (2) as indicators of reliance on centralized food production ($\mathbf{y}$) in Equation (3). Whereas labor-market signals arise from changes in the allocation of production tasks, consumer-market signals emerge from how consumers' perceptions of the food-production process, their consistent behavior within the MIMIC framework is difficult to attribute merely to a direct influence of wage structures on consumer complaints. Therefore, if the estimated factor loadings are consistent with theoretical expectations and the model's over-identifying restrictions cannot be rejected, this would support the interpretation that both indicators capture a common underlying production arrangement: restaurants' reliance on centralized food production.

$$\begin{aligned} WageRatio_i &= \lambda_1\eta_i + \nu_{1i}, \quad \lambda_1 < 0 \\ CommentRatio_i &= \lambda_2\eta_i + \nu_{2i}, \quad \lambda_2 > 0 \end{aligned} \tag{5}$$

We include *Rent*, *Chain* and *Regulation* as the vector of exogenous causal variables (Factors) as $\mathbf{x}$ in Equation (4).

$$\eta_i = \alpha_1 Rent_i + \alpha_2 Chain_i + \alpha_3 Regulation_i + \zeta_i \tag{6}$$

where $Rent$ is the daily rent per square meter of restaurant i, $Chain$ is the log of the number of outlets of the brand associated with restaurant i, to account for the potential non-linear relationship between brand scale and its reliance on prepared meal, and $Regulation$ is a dummy variable, if the location of restaurant i makes its usage of open fire become illegal, $Regulation_i = 1$, if not, $Regulation_i = 0$.

TABLE 2. Definitions of Variables

Variable	**Definition**
Indicators	
WageRatio	Ratio of the wage offered to a mid-level chef to that offered to a server at the same restaurant.
CommentRatio	Number of negative reviews explicitly mentioning prepared meals, central-kitchen production, or related production practices, divided by the total number of negative reviews.
Factors	
Chain	Natural logarithm of the total number of outlets operated by the restaurant chain.
Rent	Daily commercial rent per square meter in the restaurant's local area.
Regulation	Dummy variable, equals to 1 if the use of open flames is prohibited at the restaurant's location, and 0 otherwise.

All the factor and indicator variables used in our analysis along with their descriptions are summarized in Table 2, and the mechanism of the MIMIC model is visualized in figure 1.

[Insert Figure 1 about here]

5 Data Analysis and Findings

5.1 Data

This analysis uses data collected in 2026, our original data set contains the wage information of 351 different restaurants in five major cities in China: Guangzhou, Shenzhen, Beijing, Shanghai, and Hangzhou. The initial sample was obtained from BOSS Zhipin, one of China's major online recruitment platforms (Tang, 2016). By using a web-scraping program, we collected restaurant job postings in which the same restaurant simultaneously recruited both front-of-house servers and back-of-house kitchen workers. After removing duplicated postings, non-restaurant businesses, the final labor-market sample contains 351 restaurants.

For each restaurant, we recorded the advertised server wage and the advertised kitchen-worker wage. Audoly et al. (2024) find that, for job advertisements providing pay-related information, most of them explicitly revealing the actual salary information. Unlike countries with established tipping systems such as the United States, restaurant servers in China generally do not receive gratuities as a systematic component of compensation (Liu, 2003). Therefore, posted wages provide a closer approximation of their actual labor-market returns.

Since kitchen jobs differ substantially in skill content, we manually classified chef-related positions into three categories based on job titles and stated job requirements. Junior kitchen positions include tasks such as cutting, dish washing, and other auxiliary back-kitchen work. Middle-level chef positions refer to jobs directly responsible for cooking and dish preparation. Senior chef positions include managerial or administrative responsibilities, such as executive chef, and kitchen-management roles. The empirical analysis uses the wage of middle-level chefs to construct the wage ratio, because according to Acemoglu and Autor (2011) definition of medium skill workers, this category is most directly exposed to substitution by centralized food production. Junior kitchen workers do not fully represent skilled cooking labor, while senior chefs often combine culinary work with management functions. Middle-level chef wages therefore provide the cleanest measure of the labor-market value of restaurant-level cooking skills. Since most job postings give an interval of wage rather than a specific wage, we adopt the midpoint of the interval. This procedure was applied consistently to both server and chef positions.

Following the approach of Boeing and Waddell (2017), who utilized web-scraped online rental listings to construct fine-grained spatial measures of local market conditions, we collect commercial rental information from Anjuke, one of the largest online real estate platforms in China that provides extensive listings of commercial properties. A web-scraping program was used to obtain rental prices and geographic coordinates of commercial spaces. These rental observations were then geographically matched with restaurants identified from BOSS Zhipin to construct a restaurant-level measure of rental cost pressure.

Following prior research that measures restaurant chain scale using publicly available outlet information (e.g., company disclosures and franchise records), we constructed a chain-size measure based on the number of outlets operated by each restaurant brand (China Chain Store & Franchise Association and China Renaissance, 2022; Zhang, 2025). Since outlet information is not uniformly disclosed across Chinese restaurant brands, we adopted a hierarchical manual search procedure. We first collected outlet numbers from official brand websites. When such information was unavailable, we relied on Jiamengxing, a Chinese franchise-information platform. Finally, for brands lacking both sources, we manually retrieved store counts from Dianping.

In addition to labor-market and operating-environment data, we manually collected consumer-review information and restaurant consumption per capita from Dianping, one of the largest Chinese online restaurant review platforms, providing user-generated reviews and restaurant information that have been widely used in empirical studies of restaurant markets (Wu et al., 2015). As Dianping explicitly restricts automated scraping[11], this part of the dataset was collected manually. For each restaurant labor-market sample, we recorded per-capita consumption, the total number of reviews, the total number of negative reviews, and the number of negative reviews that explicitly criticized prepared meals, semi-finished dishes, reheated food, central-kitchen production, or similar standardized food-preparation practices. In total, we collected 24,615 negative reviews, among which 1,339 were identified as involving prepared-meal-related criticism. To reduce measurement error, Dianping reviews were independently collected by four researchers and reevaluated if detected any difference.

After excluding restaurants with missing or unreliable information, we obtained a final dataset comprising 311 restaurants. This sample size is considered adequate for estimating the MIMIC model employed in this study. With two reflective indicators and three observed causes, the reduced-form coefficient matrix is a 2×3 matrix constrained to have rank one, generating $(2-1) \times (3-1)$ testable restrictions and rendering the model over-identified (Jöreskog and Goldberger, 1975). Given the five observed variables and the two degrees of freedom, a full mean-and-covariance formulation implies approximately 18 freely estimated parameters, corresponding to an observation-to-parameter ratio of $17.3 : 1$. This ratio is substantially above the conventional minimum of approximately five observations per estimated parameter, while the total sample also exceeds the general benchmark of 150–200 observations recommended for structural equation models (Jöreskog and Goldberger, 1975). Moreover, previous applied MIMIC studies have estimated models with 2 or 3 indicators using samples of 288 (Lafuente-Ruiz-de Sabando et al., 2019) and 200 (Barrales-Molina et al., 2013). Therefore, a sample size of more than 300 observations is considered adequate for the estimation of the model.

For the external validation, we identified 21 restaurant brands with publicly documented use of centralized food production, based on either corporate disclosures or authoritative media reports. Of the 351 restaurants in the original labor-market dataset, five outlets belonged to these brands and were classified as known positives, leaving 346 restaurants in the no-public-disclosure group. We then supplemented the known-positive sample with wage information from 48 additional outlets of the identified brands. The final validation sample therefore consisted of 53 known-positive and 346 no-public-disclosure restaurants, for a total of 399 unique observations.

5.2 Findings

This Subsection is divided into four parts: Section 5.2.1 reports the descriptive statistics, Section 5.2.2 reports the results for the MIMIC model, and the robustness results are reported in Section 5.2.3. Section 5.2.4 uses data from restaurants which have already voluntarily disclosed their use of prepared meals or have been disclosed by authoritative medias to conduct external validation.

5.2.1 Descriptive statistics

Table 3 indicates the summary statistics of the wages, factors, and indicators used in the analysis. Comparing the wages of servers and chefs, the sample mean wage of chefs appears higher than that of servers, which is consistent with the traditional division of labor in restaurants where cooking tasks generally require more specialized skills. Meanwhile, the standard deviation of chef wages is substantially larger than that of server wages, which is consistent with our theoretical expectation that the adoption of centralized food production may affect different restaurant tasks asymmetrically.

[11] See Section 5 (Intellectual Property) of the Meituan User Service Agreement (Meituan, 2023), which explicitly restricts unauthorized data scraping using automated tools, including robots and web spiders.

TABLE 3. Descriptive Statistics

Variable	Mean	Median	Standard Deviation
Chef Wage	7.079	7.000	1.500
Server Wage	5.423	5.500	0.921
Indicators			
Wage Ratio	1.335	1.273	0.320
Comment Ratio	0.082	0.039	0.122
Factors			
Chain	2.277	1.386	2.338
Rent	6.417	4.670	5.534
Regulation	0.277	–	–

Note: Wage variables are measured in thousand RMB per month, but Rent is measured in RMB/m^2/day

5.2.2 Results from the MIMIC model

We present the estimates from the MIMIC model in Table 4. Results from three models are presented. The results indicate that all the factors have expected signs and significance. As shown in Table 4, the coefficients of rent, chain scale, and regulation are all positive and statistically significant at 1 percent level, indicating that higher rents, greater chain scale, and open-fire cooking restrictions are positively associated with restaurants' reliance on prepared meals.

TABLE 4. Results from MIMIC Model

Variable	Unstandardized Coefficient	Standard Error	*P*	Standardized Coefficient	95% CI
Indicators					
Comment Ratio	1(constrained)	-	-	0.575	-
Wage Ratio	−3.408***	0.419	0.000	-0.744	[−4.229, −2.586]
Factors					
Rent	0.004***	0.000	0.000	0.373	[0.003, 0.006]
Chain	0.011***	0.002	0.000	0.381	[0.007, 0.015]
Regulation	0.063***	0.011	0.000	0.408	[0.041, 0.086]
Diagnostic statistics	**Value**				
Observations	311				
χ^2 [p-value]	3.307 [0.191]				
CD	0.611				
CFI	0.994				
TLI	0.978				
RMSEA	0.046				
SRMR	0.018				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

Diagnostic statistics of the estimated MIMIC model is reported in Table 4. A necessary condition for model identification is: $(p \times q + 1/2(p)(p+1) - 2p - q) \geq 0$ where p denotes the number of indicator variables and q is the number of causal variables. According to Jöreskog and Goldberger (1975), the likelihood-ratio test jointly examines whether the reduced-form coefficient matrix satisfies the rank-one restriction implied by the MIMIC model and whether the residual covariance matrix is consistent with a common latent factor and indicator-specific measurement errors. Under the null hypothesis this test

statistic is distributed as a χ^2 with $q(p-1)+\frac{1}{2}p(p-3)$ degrees of freedom. In our model, $p=2$ and $q=3$, yielding two degrees of freedom. The rank-one restriction is substantively important for the present study because it requires the effects of the observed factors on the two indicators to be proportional to a common loading vector (Jöreskog and Goldberger, 1975). In other words, *Rent*, *Chain*, and *Regulation* cannot generate arbitrary patterns of association with *WageRatio* and *CommentRatio* while remaining consistent with the specified MIMIC structure. Rather, the model tests whether the joint relationships between the observed causes and indicators conform to the restrictions implied by a common latent construct, without requiring every conceivable direct pathway to be ruled out individually.

The second row of diagnostic statistics in Table 4 reports the result from the over-identification test. Rejection of the null hypothesis would suggest the estimated model is mis-specified. $\chi^2 = 3.307$ with its $p = 0.191$, as stated row 2 of Table 4, clearly indicate that the estimated model is not mis-specified. Given this evidence, we then provide a series of model fit diagnostics to render the estimated model statically adequate. We use five different statistics: the coefficient of determination (CD), the comparative fit index (CFI), the Trucker-Lewis index (TLI), the root mean square of approximation (RMSEA) and the standardized root mean square of residual (SRMR). These statistics indicate a measure of overall good-fit of our estimated model.

[Insert Figure 2 about here]

5.2.3 Robustness check

Since the reliability of indicator *CommentRatio* may be compromised for restaurants with only a small number of reviews, we exclude restaurants with fewer than 100 total reviews and re-estimate the model using the restricted sample. The results of this model is depicted by Table 5 and visualized in Figure 2. The results remain consistent.

TABLE 5. Results from MIMIC model excluding all the restaurants with reviews less than 100

Variable	**Unstandardized Coefficient**	**Standard Error**	*P*	**Standardized Coefficient**	**95% CI**
Indicators					
Comment Ratio	1(constrained)	-	-	0.615	-
Wage Ratio	-3.750^{***}	0.505	0.000	-0.730	[−4.741, −2.760]
Factors					
Rent	0.004^{***}	0.000	0.000	0.333	[0.002, 0.006]
Chain	0.010^{***}	0.002	0.000	0.392	[0.006, 0.015]
Regulation	0.058^{***}	0.011	0.000	0.437	[0.037, 0.080]
Diagnostic statistics	**Value**				
Observations	234				
χ^2 [*p*-value]	3.888 [0.143]				
CD	0.608				
CFI	0.989				
TLI	0.960				
RMSEA	0.064				
SRMR	0.022				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

However, a restaurant with 20 prepared-meal-related reviews out of 40 negative reviews and a restaurant with only 1 such review out of 2 negative reviews would have same *CommentRatio*. Nevertheless, the former provides substantially stronger evidence of reliance on centralized food production. To address

this limitation caused by the structure of $CommentRatio$, we replace $CommentRatio$ in Equation (5) with N^{PM} which denotes the absolute number of negative reviews that explicitly mention prepared meals. The result of this model is depicted in Table 6.

TABLE 6. Robustness Check: Numbers of Comments mentioned Prepared Meals

Variable	**Unstandardized Coefficient**	**Standard Error**	***P***	**Standardized Coefficient**	**95% CI**
Indicators					
N^{PM}	1(constrained)	-	-	0.597	-
Wage Ratio	-0.057^{***}	0.006	0.000	-0.639	[−0.070, −0.045]
Factors					
Rent	0.294^{***}	0.040	0.000	0.454	[0.215, 0.373]
Chain	0.738^{***}	0.106	0.000	0.481	[0.530, 0.946]
Regulation	3.397^{***}	0.529	0.000	0.425	[2.360, 4.434]
Diagnostic statistics	**Value**				
Observations	311				
χ^2 [p-value]	1.110 [0.574]				
CD	0.828				
CFI	1.000				
TLI	1.013				
RMSEA	0.000				
SRMR	0.010				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

The alternative specification remains excellent model fit, and the signs and statistical significance of the measurement and structural coefficients remained consistent with those of the baseline model.

To address potential heterogeneity across restaurant formats, we conduct an additional robustness check by excluding fast-food and hotpot restaurants. These two categories may exhibit inherently different production structures: fast-food restaurants often rely on standardized production systems, while hotpot restaurants require relatively limited cooking labor (China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute, 2023). Therefore, including them may mechanically reduce the $WageRatio$ independent of centralized food production reliance. After exclusion, the estimated relationships remain stable. According to Table 7, the model remains acceptable after excluding fast-food and hotpot restaurants, with the likelihood-ratio test failing to reject the model specification. All the factors remain positively associated with latent reliance on centralized food production. Moreover, most of the diagnostics statistics continue to perform well. Although RMSEA increases to 0.100, this measure can be unstable in models with low degrees of freedom and small sample (Jöreskog and Goldberger, 1975).

One potential concern is that the wage ratio may simply reflect differences in restaurants' market positioning or price levels. A potential alternative explanation is that high-end restaurants pay relatively higher wages to front-of-house servers because they require more intensive service, more formal dining procedures, and better customer interaction. Under this interpretation, a lower chef-to-server wage ratio may simply reflect higher server wages in expensive restaurants, rather than a reduction in skilled back-kitchen labor caused by prepared-meal adoption.

To address this concern, we use restaurant-level consumption per capita as a placebo outcome. If the wage ratio primarily reflects restaurant quality, luxury positioning, or high-end service intensity, it should be systematically associated with consumption per capita. In particular, if expensive restaurants have relatively higher server wages, then the chef-to-server wage ratio should be negatively associated with

TABLE 7. Robustness Check: MIMIC Model Excluding Fast-Food and Hot-Pot Restaurants

Variable	Unstandardized Coefficient	Standard Error	P	Standardized Coefficient	95% CI
Indicators					
Comment Ratio	1(constrained)	-	-	0.652	-
Wage Ratio	-3.588^{***}	0.512	0.000	-0.694	[−4.592, −2.583]
Factors					
Rent	0.0039^{***}	0.0008	0.000	0.336	[0.002, 0.006]
Chain	0.0112^{***}	0.0024	0.000	0.405	[0.006, 0.016]
Regulation	0.0611^{***}	0.0117	0.000	0.447	[0.038, 0.084]
Diagnostic statistics	**Value**				
Observations	196				
χ^2 [p-value]	5.939 [0.051]				
CD	0.638				
CFI	0.973				
TLI	0.904				
RMSEA	0.100				
SRMR	0.030				

Note:
** Denotes significance at 5 percent level.
*** Denotes significance at 1 percent level.

consumption per capita. By contrast, if the wage ratio captures back-kitchen deskilling associated with prepared-meal reliance, rather than general restaurant price level, its relationship with consumption per capita should be weak or statistically insignificant.

Formally, we estimate the following placebo outcome regression:

$$ConsumptionPerCapita_i = \theta_0 + \theta_1 WageRatio_i + \varepsilon_i,$$

where $ConsumptionPerCapita_i$ denotes the average spending per customer at restaurant i.

As reported in Table 8, the estimated coefficient of Wage Ratio is statistically insignificant ($p = 0.592$), indicating that the plausibility that the indicator simply reflects restaurant tier rather than differences in production practices is reduced.

TABLE 8. Placebo Test: Wage Ratio and Consumption Level

Variable		Coefficient	Standard Error	P	95% CI
Wage Ratio		12.314	22.923	0.592	[−32.790, 57.419]
Constant		97.172	31.459	0.002	[35.271, 159.074]
Diagnostic Statistics	Observations	R^2	Adjusted R^2	F-statistics	p-value
Value	311	0.001	−0.002	0.29	0.592

Note: Standard errors are reported in parentheses. The dependent variable is consumption per capita.

5.2.4 External Validation of the Wage-Ratio Indicator

Although the MIMIC model provides internal construct-validity evidence that the $WageRatio$ behaves consistently with the latent reliance on centralized food production, the interpretation of this indicator can be further strengthened through external validation. The purpose of this section is to examine whether restaurants that actively or passively admit their use of centralized food production exhibit the theoretically expected wage-structure pattern.

We construct a validation group based on restaurants' official website or authoritative media's report. A restaurant is classified into the known-positive group if official websites, franchise documents, prospectuses, annual reports, media interviews, or other public materials explicitly disclose that the restaurant adopts a central-kitchen production system or relies on centrally prepared meal components.

Let C denote the set of restaurants that publicly disclose the use of central-kitchen or prepared-meal production. We define:

$$CentralKitchen_i = \begin{cases} 1, & i \in C \\ 0, & i \notin C \end{cases}$$

It should be emphasized that $CentralKitchen_i = 0$ does not imply that restaurant i does not use centralized production. It only indicates that no public disclosure of the use of centralized food production is available. Therefore, the validation sample should be interpreted conservatively: restaurants with $CentralKitchen_i = 1$ serve as a known-positive group, while the remaining restaurants are not treated as confirmed non-users.

As shown in Table 9, group C has a significantly lower mean $WageRatio$ than the no-public-disclosure group. The difference is both statistically significant ($p < 0.001$) and large in magnitude (Cohen's $d = -1.097$), providing external support for the validity of the wage-ratio indicator.

TABLE 9. External Validation of the Wage-Ratio Indicator

Group	**Observations**	**Mean**	**Standard Deviation**	**Standard Error**
Known-positive group (*CentralKitchen* = 1)	53	1.027	0.069	0.009
No-public-disclosure group (*CentralKitchen* = 0)	346	1.347	0.311	0.017
Between-group comparison	**Value**			
Mean difference	-0.319			
Welch's t	-16.635			
Degrees of freedom	358.912			
p-value	< 0.001			
95% CI	$[-0.357, -0.282]$			
Cohen's d	-1.097			
95% CI for Cohen's d	$[-1.395, -0.797]$			

Note: The original dataset consists of 351 restaurants. Among them, five restaurants were identified as the Known-positive group based on publicly disclosed information regarding central kitchen or prepared-meal usage. The remaining 346 restaurants constitute the no-public-disclosure group. Mean differences are calculated as $CentralKitchen = 1$ minus $CentralKitchen = 0$. Welch's t-test is reported because Levene's test rejects the equal-variance assumption (F=48.202,p¡0.001).

To further address the concern that the external validation results may be driven by multiple outlets from the same brand, we conduct an additional robustness check using a brand-level de-duplicated sample. Specifically, for brands with multiple publicly disclosed outlets, only one outlet is retained to ensure that each brand contributes only one observation. The resulting sample contains 21 known-positive restaurants without brand duplication.

As shown in Table 10, the results remain consistent, this confirms that the observed difference is not driven by the over-representation of particular brands in the validation sample.

TABLE 10. Robustness Test of External Validation

Group	Observations	Mean	Standard Deviation	Standard Error
Known-positive group ($CentralKitchen = 1$)	21	1.022	0.075	0.016
No-public-disclosure group ($CentralKitchen = 0$)	346	1.347	0.311	0.017
Between-group comparison	**Value**			
Mean difference	-0.325			
Welch's t	-13.897			
Degrees of freedom	78.882			
p-value	< 0.001			
95% CI	$[-0.371, -0.278]$			
Cohen's d	-1.072			
95% CI for Cohen's d	$[-1.519, -0.624]$			

Note: Mean differences are calculated as $CentralKitchen = 1$ minus $CentralKitchen = 0$. Welch's t-test is reported because Levene's test rejects the equal-variance assumption ($F = 19.457$, $p < 0.001$).

6 Discussion and Conclusions

6.1 Discussion

The central finding of this study lies in the fact that the hidden production practice of restaurants can be investigated through signals from distinct market. Results from the MIMIC model indicate that H1-H5 were to supported by the data. The association between $WageRatio$, $CommentRatio$ and centralized food production reliance are consistent with our intuition.

Moreover, this study empirically confirms that the chain size, rent, and fire regulation can affect a restaurant's centralized production reliance in positive direction. Higher commercial rent may push the restaurant to apply centralized production, as it can effectively reduce the necessary kitchen area. Moreover, The chain size has positive effect on the use of centralized food production due to the pressure to unify flavor across different stores of a brand. Besides, strict fire-safety regulation in large commercial complexes may also lead to higher reliance to centralized food production.

Several robustness analyses further assess whether the observed relationships are driven by alternative indicator construction, restaurant heterogeneity, or differences in market positioning (see figure 3). Beyond the internal consistency of the MIMIC model, external validation provides an additional source of evidence by comparing wage structures with independently documented production practices (see section 5.2.4).

[Insert Figure 3 about here]

Taken together, the convergence of theoretical predictions, model-based evidence, robustness analyses, and external validation supports the proposed measurement interpretation, while leaving room for further validation of strong causal mechanisms.

6.2 Contributions to Research

The broad objective of this study was to further our appreciation of factors that motivate a restaurant to adopt centralized production. Therefore, the key distinguishing facet of the study concerns the fact that it empirically discussed the factors that can bear down a restaurant's decision to centralized production. As noted in Section 1, no prior empirical study offering such a systematic listing of causal factors, including perspectives from operating cost, standardization demand, economies of scale and institutional regulation, influencing centralized production decision has been found in the literature, since the production mode of restaurants is hard to be observed directly. The MIMIC results reveal that

restaurants' reliance on centralized food production is associated not only with the pursuit of operational efficiency but also with broader economic and institutional conditions, including local operating costs, regulatory constraints, and chain expansion. These findings broaden our understanding of production centralization as an organizational choice shaped by both local operating environments and the potential benefits of standardization and economies of scale in chain operations.

A second contribution of this study is its extension of the task-based framework proposed by Acemoglu and Autor (2011). As noted in Section 2.3, most applications of this model stressed on using this model to explain or predict the change in relative wage or inequity of income under the impact of technological and offshoring impact. This study apply this model reversely, by observing difference in relative wage, we can identify the differe in the production organization, which is unobservable originally. This extends the application of task-based model and can be generalized to other sectors for identifying the hidden characteristics of production mode.

A third contribution of this study is to provide a new perspective on identifying unobservable production practices through information generated in adjacent markets. Rather than treating labor information solely as an indicator of worker welfare, we demonstrate that it can also serve as a diagnostic signal of otherwise unobservable production practices. This study illustrates this possibility by using labor-market information, originally generated for recruitment purposes, to infer restaurants' otherwise unobservable reliance on centralized food production. By incorporating such information into a latent-variable framework, the study demonstrates how observable signals from adjacent markets can provide indirect evidence of hidden production arrangements. More broadly, this finding suggests that opaque production practices in one market may be inferred from observable economic traces in another, offering a potentially generalizable approach to investigating production organization in sectors where direct observation is limited.

6.3 Contributions to Practice

Our research also contributes to managerial practice in various ways. It allows easier assessment of reliance on centralized food production as it lowers the requirements for data collection. By utilizing existing data (as we do by referring data from public recruitment platforms and review websites), we are capable of assessing a restaurant's reliance on centralized production. This approach may help managers benchmark competitors and better understand the operational practices adopted by other restaurants in their market.

Further, by showing the positive relationship between rental cost, chain sizeinstitutional regulation and reliance on centralized food production empirically, we discover that restaurants' adoption of centralized production is not merely a transition in production process, but a organizational strategic choice, when considering the adoption of centralized production, managers should not only care about the processing cost of food, regulation constraint, rental cost and pressure of chain's management should also be taken into consideration.

Second, this study provides implications for restaurant labor management and the allocation of tasks between front-of-house (FOH) and back-of-house (BOH) operations. Since substantial culinary tasks are transferred to centralized facilities, managers should reassess BOH staffing, skill requirements, and training accordingly. Moreover, as the adoption of centralized production does not necessarily reduce customer-facing service requirements, managers should ensure effective FOH–BOH coordination while balancing production standardization against culinary differentiation and service quality.

Third, the findings highlight the importance of production transparency and the consistency between restaurant production practices and brand positioning by proving the positive relationship between $CommentRatio$ and reliance on centralized food production. Although centralized production may offer operational advantages, consumers may hold different expectations regarding food preparation methods, particularly in restaurants characteristic with freshly prepared dishes or traditional culinary craftsmanship. Discrepancies between actual production arrangements and consumer expectations

may consequently generate dissatisfaction and reputation crisis. Restaurant managers should therefore consider how production arrangements are communicated to consumers and ensure that claims about food preparation remain consistent with actual practices.

Besides management insights, in terms of consumers' choice, the $WageRatio$ method proposed in this study may be a practical guide for consumers. As with other indicators or signals, measuring restaurant reliance on prepared meals is ambiguous and unreliable. Restaurants do not have the incentive to actively disclose their reliance on prepared meals. The chef-to-server wage ratio may serve as a supplementary signal for consumers to identify restaurants that may highly rely on prepared meals . Though a restaurant's reliance on prepared meals cannot be simply determined by wage ratio, we believe that consumers can refer to $WageRatio$ as an estimation if they do mind having prepared meals in restaurants.

6.4 Limitations and Suggestions for Future Research

As with any research, this study is not without limitations. Due to data availability constraints, we rely on cross-sectional data collected within a single period rather than panel data tracking restaurants over time. Although the cross-sectional design allows us to identify systematic differences across restaurants, it cannot fully capture changes in restaurants' production practices over time. Moreover, the observational nature of the data limits our ability to establish strong causal relationships between operational factors and reliance on centralized food production.

A further limitation concerns the generalization of the proposed measurement framework. The methodology employed in this study is partly grounded in the cultural characteristics of East Asian restaurant industry, particularly the limited prevalence of tipping and consumers' sensitivity to prepared meals. These characteristics may differ across countries and restaurant markets, potentially affecting the applicability of the proposed indicators beyond the East Asian context.

Future research could construct panel datasets by continuously tracking restaurants' labor-market signals, consumer reviews, and operational characteristics over multiple periods. Such longitudinal data would enable researchers to examine how restaurants' reliance on centralized food production evolves over time and provide opportunities for more rigorous causal analysis. Furthermore, combining longitudinal observations with independently verified information on restaurants' production practices could facilitate the identification of appropriate classification thresholds and the evaluation of the sensitivity and specificity of the proposed indicators, thereby enhancing their practical applicability.

References

Daron Acemoglu and David Autor. Skills, tasks and technologies: Implications for employment and earnings. In *Handbook of Labor Economics*, volume 4, pages 1043–1171. Elsevier, 2011. ISBN 978-0-444-53452-1.

R. Audoly, M. Bhuller, and T. A. Reiremo. The pay and non-pay content of job ads. Staff Report 1124, Federal Reserve Bank of New York, 2024. URL https://doi.org/10.59576/sr.1124.

David H. Autor, Frank Levy, and Richard J. Murnane. The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics*, 118(4):1279–1333, 2003. doi:10.1162/003355303322552801.

Li Bai, Mingliang Wang, Yinsheng Yang, and Shunlong Gong. Food safety in restaurants: The consumer perspective. *International Journal of Hospitality Management*, 77:139–146, 2019. ISSN 0278-4319. doi:10.1016/j.ijhm.2018.06.023. URL https://www.sciencedirect.com/science/article/pii/S0278431917305650.

Vanesa Barrales-Molina, Oscar F. Bustinza, and Leopoldo J. Gutiérrez-Gutiérrez. Explaining the causes and effects of dynamic capabilities generation: A multiple-indicator multiple-cause modelling approach. *British Journal of Management*, 24(4):571–591, 2013. doi:10.1111/j.1467-8551.2012.00829.x.

Alan Blinder. How many u.s. jobs might be offshorable? Working Paper 142, Princeton University Center for Economic Policy Studies, 2007.

Alan Blinder and Alan B. Krueger. Measuring offshorability: A survey approach. Working paper, Princeton University, 2008.

Geoff Boeing and Paul Waddell. New insights into rental housing markets across the united states: Web scraping and analyzing craigslist rental listings. *Journal of Planning Education and Research*, 37(4): 457–470, 2017.

Jeffrey L. Bradach. *Franchise Organizations*. Harvard Business School Press, Boston, MA, 1998.

Jaemun Byun and SooCheong Jang. Open kitchen vs closed kitchen. *International Journal of Contemporary Hospitality Management*, 30(5):2214–2229, 2018. doi:10.1108/IJCHM-03-2016-0167.

Yong Chen and Kai Jiang. A multiple indicators multiple causes (MIMIC) model of the behavioral consequences of hotel guests. *Tourism Management Perspectives*, 30:197–207, 2019. doi:10.1016/j.tmp.2019.02.014.

China Chain Store & Franchise Association and China Renaissance. 2022 china chain catering industry report. Industry report, China Chain Store & Franchise Association, 2022.

China Cuisine Association and Shuhai Supply Chain and Baichuan Research Institute. 2023 china catering supply chain white paper [2023zhongguo canyin gongyinglian baipishu]. White paper, Shuhai Supply Chain, 2023. URL https://www.shuhaisc.com. Accessed July 6, 2026.

Pierre-Philippe Combes, Gilles Duranton, and Laurent Gobillon. Spatial wage disparities: Sorting matters! *Journal of Urban Economics*, 63(2):723–742, 2008. doi:10.1016/j.jue.2007.04.004.

Ai Deng. Supply chain innovation in the catering industry: An assessment of the cost-effectiveness and sustainability of the central kitchen model. *Advances in Economics, Management and Political Sciences*, 65:188–194, 2023. doi:10.54254/2754-1169/65/20231630.

Lu Dong, Caiyi Qiu, Fan Wei, Zhenting Yu, Yan Zhang, and Shuo Wang. The effect of microwave baking conditions on the quality of biscuits and the control of thermal processing hazards in the maillard reaction. *Frontiers in Nutrition*, 9:825365, 2022. doi:10.3389/fnut.2022.825365.

Milena Flessas, Vinícius Rizzardi, Guilherme Tortorella, Diego Fettermann, and Giuliano Marodin. Layout performance indicators and systematic planning: A case study in a southern brazilian restaurant. *British Food Journal*, 117(8):2078–2097, 2015. doi:10.1108/BFJ-01-2015-0012. URL https://doi.org/10.1108/BFJ-01-2015-0012.

Bruno S. Frey and Hannelore Weck-Hanneman. The hidden economy as an "Unobserved" variable. *European Economic Review*, 26(1):33–53, 1984. doi:10.1016/0014-2921(84)90020-5.

Merve Göde and Vedat Ekergil. Designing an activity-based costing method for food and beverage businesses: A determination of activities. In *Advances in Service Management and Social Media*, pages 101–112. Springer, 2020. ISBN 978-981-10-7067-9. doi:10.1007/978-981-10-7068-6_6.

Alenica Hässig, Christina Hartmann, Luisma Sanchez-Siles, and Michael Siegrist. Perceived degree of food processing as a cue for perceived healthiness: The NOVA system mirrors consumers' perceptions. *Food Quality and Preference*, 110:104944, 2023. ISSN 0950-3293. doi:10.1016/j.foodqual.2023.104944. URL https://www.sciencedirect.com/science/article/pii/S0950329323001386.

IBISWorld. Chain restaurants in the us - market research report, 2024. URL https://www.ibisworld.com/united-states/market-research-reports/chain-restaurants-industry/. Retrieved from IBISWorld database.

Yangyang Jia, Lanlan Hu, Ruifeng Liu, Wei Yang, Ibrahim Khalifa, Jicai Bi, Yunbo Li, Jianlong Zhen, Beiping Wang, Zhao Zhang, Erpeng Zhang, and Bo Li. Innovations and challenges in the production of prepared dishes based on central kitchen engineering: A review and future perspectives. *Innovative Food Science & Emerging Technologies*, 91:103521, 2024. doi:10.1016/j.ifset.2023.103521.

Karl G. Jöreskog and Arthur S. Goldberger. Estimation of a model with multiple indicators and multiple causes of a single latent variable. *Journal of the American Statistical Association*, 70(351):631–639, 1975. doi:10.2307/2285946.

Hungtang Ko and David L. Hu. The physics of tossing fried rice. *Journal of the Royal Society Interface*, 17(163):20190622, 2020. doi:10.1098/rsif.2019.0622.

Amaia Lafuente-Ruiz-de Sabando, Javier Forcada, and Pilar Zorrilla. The university image: A model of overall image and stakeholder perspectives. *Cuadernos de Gestión*, 19(1):63–86, 2019. doi:10.5295/cdg.160720al.

Woojin Lee, SooCheong (Shawn) Jang, and Hong Soon Kim. The effect of digital transformation: Boosting productivity in the restaurant industry. *International Journal of Hospitality Management*, 123:103896, 2024. doi:10.1016/j.ijhm.2024.103896. URL https://doi.org/10.1016/j.ijhm.2024.103896.

H. Liu, L. Zhang, and Y. Wang. Supply chain integration and the development of central kitchens in china's restaurant sector: A sustainability perspective. *Sustainability*, 14(9):5421, 2022. doi:10.3390/su14095421.

Ying Liu. *Factors Affecting Tip Size in Restaurants: A Study of Chinese Restaurants in Hong Kong*. PhD thesis, The Hong Kong Polytechnic University, 2003.

J. Kenji López-Alt. Wok skills 101: Stir-frying basics, 2012. URL https://www.seriouseats.com/wok-skills-101-stir-frying-basics. Serious Eats. Accessed July 12, 2026.

Meituan. Meituan user service agreement. Online terms of service, December 2023. URL https://rules-center.meituan.com/m/detail/4. Published December 28, 2023; accessed October 15, 2026.

Ministry of Emergency Management of the People's Republic of China. Fire safety management rules for large commercial complexes. Technical Report XF/T 3019-2023, Ministry of Emergency Management of the People's Republic of China, July 2023. Implemented October 19, 2023.

Mary Frances Nettles, Mary B. Gregoire, and Deborah D. Canter. Analysis of the decision to select a conventional or cook-chill system for hospital foodservice. *Journal of the American Dietetic Association*, 97(6):626–631, 1997. doi:10.1016/S0002-8223(97)00159-4.

Sorcha O'Neill, Mark Traynor, Imran Rahman, and Yee Lee. Exploring front-of-house and back-of-house manager perceptions, attitudes, and motivations of restaurant food waste mitigation: A qualitative study. *Sustainability*, 16(15):6694, 2024. doi:10.3390/su16156694. URL https://doi.org/10.3390/su16156694.

Emmanuel Purlis. Browning development in bakery products–a review. *Journal of Food Engineering*, 99 (3):239–249, 2010. doi:10.1016/j.jfoodeng.2010.03.008.

Richard N. S. Robinson and Paul Barron. Developing a framework for understanding the impact of deskilling and standardisation on the turnover and attrition of chefs. *International Journal of Hospitality Management*, 26(4):913–926, 2007. doi:10.1016/j.ijhm.2006.10.002.

Svetlana Rodgers. Selecting a food service system: A review. *International Journal of Contemporary Hospitality Management*, 17(2):147–156, 2005. doi:10.1108/09596110510582341.

Joel Mark P. Rodriguez, Hanna V. Reambonanza, and Lourdes Q. Palallos. Optimizing operation processes and supply chain management for enhanced service and product quality in quick service restaurants. *The South East Asian Journal of Management*, 19(1):100–122, 2025. doi:10.7454/seam.v19i1.1788.

Scott E. Sampson. A strategic framework for task automation in professional services. *Journal of Service Research*, 24(1):122–140, 2021. doi:10.1177/1094670520940407.

Haiyan Shan, Yuan Dai, and Wenjie Pi. Mitigating consumer boycott behavior in the prepared dishes industry: a tripartite evolutionary game perspective. *Frontiers in Sustainable Food Systems*, 10:1794264, 2026. doi:10.3389/fsufs.2026.1794264. URL https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2026.1794264/full.

Tao Shu, Han Yang, Ling Lin, Jian Chen, Jixian Zhou, and Jun Wang. Exploring public opinion on health effects of prepared dishes in china through social media comments. *Frontiers in Public Health*, 12: 1424690, 2024. doi:10.3389/fpubh.2024.1424690. URL https://doi.org/10.3389/fpubh.2024.1424690.

State Administration for Market Regulation. Official interpretation of the notice on strengthening food safety supervision and promoting high-quality development of prepared dishes, 2024. URL https://www.samr.gov.cn/zw/zfxxgk/fdzdgknr/xwxcs/art/2024/art_0689017d44944810a6300f3d3da33a4b.html.

Gülüm Sumnu. A review on microwave baking of foods. *International Journal of Food Science & Technology*, 36(2):117–127, 2001. doi:10.1046/j.1365-2621.2001.00479.x.

Yingying Tang. BOSS Zhipin: Embracing recruitment 2.0. *Business Observer*, (2):54–56, 2016.

Mike G. Tsionas, A. George Assaf, and Florian Kock. Measuring managerial ability in the hotel industry. *International Journal of Hospitality Management*, 103:103223, 2022. doi:10.1016/j.ijhm.2022.103223.

Tungee Research Institute. 2022 prepared meals industry research report: Navigating a rapidly expanding market. Industry report, Tungee Research Institute, July 2022. URL https://cdn.tungee.com/upload/official-website/%E9%A2%84%E5%88%B6%E8%8F%9C%E8%A1%8C%E4%B8%9A%E7%A0%94%E7%A9%B6%E6%8A%A5%E5%91%8A_1666748767505.pdf.

Haiyan Wang, Tava Lennon Olsen, and Guiqing Liu. Service capacity competition with peak arrivals and delay sensitive customers. *Omega*, 77:80–95, 2018. doi:10.1016/j.omega.2017.06.001. URL https://doi.org/10.1016/j.omega.2017.06.001.

Chunhua Wu, Hai Che, Tat Y. Chan, and Xianghua Lu. The economic value of online reviews. *Marketing Science*, 34(5):739–754, 2015.

Huidan Xue, Ziling He, Qiai Wang, and Wen-Hao Su. From controversy to consensus: How consumers and policy interactions shape prepared dishes and food discourse in china. *Frontiers in Sustainable Food Systems*, 9, 2025. doi:10.3389/fsufs.2025.1666257.

Ying-Jie Yu, Sheng-Ping Yang, Ting Lin, Yun-Fang Qian, Jing Xie, and Changli Hu. Effect of cold chain logistic interruptions on lipid oxidation and volatile organic compounds of salmon (*Salmo salar*) and their correlations with water dynamics. *Frontiers in Nutrition*, 7:155, 2020. doi:10.3389/fnut.2020.00155.

Y. Zhang. Improving food quality consistency of chain restaurants with central kitchen. *Modern Supply Chain Research and Applications*, 7(4):515–538, 2025. doi:10.1108/MSCRA-04-2025-0023.

Tai Zhou. In-depth report on china's prepared food industry: The industry reaches a turning point and raw-material enterprises are poised to break through. Industry research report, Essence Securities, 2021. URL https://28075254.s21i.jzaliusr.com/61/ABUIABA9GAAg8q7EkwYo572gwAc.pdf. Originally published in Chinese.

Author Contributions

Eric Wu: Conceptualization; Methodology; Formal analysis; Investigation; Data curation; Writing – original draft.
Vincent Lau: Conceptualization; Methodology; Investigation; Visualization; Data curation; Writing – original draft.
Kane Wu: Investigation; Data Curation; Validation; Writing – review and editing.
Colin Sheng: Investigation; Data curation; Validation; Writing – review and editing.
Uchenna Cyril EZE: Supervision; Project administration; Writing – review and editing.

All authors reviewed and approved the final version of the manuscript.

Acknowledgments

The authors gratefully acknowledge Jacky Yang, Summer Sun, Julina Wu, Steve Peng, Peter Huang, Iris Chen and Morina Mei for assistance with data collection and data cleaning. The authors are particularly grateful to Professor Lawrence Khoo (Chinese University of Hong Kong, Shenzhen) and Professor Hao Shi (Beijing Normal–Hong Kong Baptist University) for their professional guidance throughout the manuscript-preparation process.